

### 8.3 Concerning Reproducibility

In future iterations, we believe it would be useful to show students more examples of what a high-quality reproducibility paper looks like and explain in-depth why it is high-quality. These could be papers that were previously accepted to the MLRC, or papers from other reproducibility efforts outlined in Section 3. We want the students to understand what makes a paper a good (reproducibility) paper, that is, it has a set of (reproducibility) claims, it argues for these claims, and shows evidence to support these claims.

## 9 Conclusion

In this paper, we share our setup for the FACT-AI course at the University of Amsterdam, which teaches FACT-AI topics through reproducibility. The course set out to give students (i) an understanding of FACT-AI topics, (ii) an understanding of algorithmic harm, (iii) familiarity with recent FACT-AI methods, and (iv) an opportunity to reproduce FACT-AI solutions, through a combination of lectures, paper discussion sessions and a reproducibility project.

Through their projects, our students engaged with the open-source community by creating a public code repository (in the 2019–2020 iteration), as well as with the research community via successful submissions to the MLRC challenge (in the 2020–2021 iteration). We also detail how the 2020–2021 iteration brought about its own unique set of challenges due to the COVID-19 pandemic.

In this course, we illustrate that reproducibility is not only paramount to good science in general, but is also a fundamental component of FACT-AI. We received very positive feedback on teaching FACT-AI topics through reproducibility. We believe this was an excellent fit for our students, which not only helped motivate them for the duration of the course, but also helped them develop skills that will be essential in their future research careers, whether in the private or public sector. To generalize this course setup to other scientific domains, we suggest identifying where the lack of reproducibility in this domain area is coming from and centre the project around evaluating this component.

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Table 2: Feedback from students (a) and the MLRC (b, c).

#### (a) Feedback from students

- “Reproducing an article was hard and intensive but a really good experience.”
- “Replicating another study, seeing how (poorly) other research is performed was really eye-opening.”
- “Reproducing a paper: I believe this is a good thing to do and is an important part of academia.”
- “Gave good insights into the trustworthiness of research papers, which is apparently not great.”
- “I appreciate the critical view I have developed on papers as a result of this course. Normally I would easily accept the content of a paper, but I will be more critical from now on, as many papers are not reproducible.”
- “I think it’s really good that we get some practical insights into reproducing results from other papers, not all papers are as good as they seem to be.”
- “I really appreciated that this was the first course where students are judging state-of-the-art AI-models. In other words, students were able to experience the scientific workflow of AI.”

#### (b) Positive feedback from the MLRC

- “The report reveals a lot of dark spots of the original paper.”
- “Good reviews, strong reproducibility report, provides code reimplementation from scratch which is a strong contribution.”
- “The discussion section is a great reference point for future work.”
- “The additional experimentation is rather impressive and the report reflects an intuitive understanding of concepts such as coverage, correctness, and counterfactual explanations.”
- “The report provides good insights on how the experiments in the original paper actually work, while also generating new hypothesis to be tested for future research, which is a positive outcome.”

#### (c) Negative feedback from the MLRC

- “My main concern is that it remains unclear why some of the results are so far off from the original paper? I would have expected the authors to dig deeper on that.”
- “The paper is generally difficult to follow. The paper reads closer to an outline than a finished report.”
- “It doesn’t go above and beyond the reproduction and does not offer novel insights into the workings of the original paper.”
- “The submission failed at reproducing the original results. It is unclear whether this is due to a difference in the experimental setup or due to implementation errors.”
- “The paper reads closer to an outline than a finished report. I would encourage the authors to spend some additional time on organization, making sure that the key takeaways are made plain and that the report reads fluidly throughout.”

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